Revision | **Covers** 4-1 → 4-5 | **File** 9 / 11

TR

The 14 traps that cost marks

الأخطاء الأربعة عشر التي تضيّع الدرجات

Each of these has a specific home in this chapter. Read the list the night before the exam.

#	THE TRAP	WHAT TO DO INSTEAD
1	Confusing **mutually exclusive** with **independent**.	Mutually exclusive = no overlap ($A \cap B = \varnothing$). Independent = no influence ($P(A \cap B) = P(A)P(B)$). Two different questions, two different tests.
2	Using Addition Rule 1 when the events overlap.	Always test the intersection first. When in doubt use Rule 2; it reduces to Rule 1 automatically.
3	Reading "or" as exclusive.	In probability, "A or B" always means A, or B, *or both*.
4	Missing "**without replacement**".	Circle it. It changes every denominator after the first draw and switches you from Rule 1 to Rule 2.
5	Putting the wrong thing in the conditional denominator.	The *condition* is the denominator. $P(B\|A)$ divides by $P(A)$, never by $P(B)$.
6	Treating "of these clients..." / "of those who..." as unconditional.	That phrasing always hands you a conditional probability. Annotate it as $P(A\|H)$ before you calculate.
7	Mixing up "at most" and "less than" (and "at least" / "more than").	**at** → the boundary value is *included*. **than** → the boundary value is *excluded*.
8	Doing "at least one" the long way.	$1 - P(\text{none})$. Always.
9	Using a permutation for a committee.	A committee has no internal order → combination. Labelled roles (chair, secretary) → permutation.

#	THE TRAP	WHAT TO DO INSTEAD
10	Adding instead of multiplying across categories.	"3 women **and** 2 men" → $_7C_3 \times {}_5C_2$. "And" is multiplication.
11	Writing 0! = 0.	$0! = 1$, by definition. It appears in $_nP_n$ and $_nC_n$.
12	Mixing permutations and combinations between $n(E)$ and $n(S)$.	Count both with the same rule. Consistency, not cleverness.
13	Giving only the final number.	Write the sample space in set notation, name the rule you are using, show the substitution. The lecturer states explicitly that set notation is required.
14	Reporting a probability greater than 1 or negative.	$0 \leq P \leq 1$. If your answer falls outside, you have added when you should have multiplied, or forgotten to divide by $n(S)$. Catch it before you move on.

الأخطاء الأربعة عشر بالعربي

1. الخلط بين **متنافيين** (لا تداخل) و**مستقلين** (لا تأثير).
2. الجمع المباشر مع أحداث متداخلة — اطرح التقاطع.
3. فهم **or** على أنها «أحدهما فقط» — هي «أحدهما أو كلاهما».
4. تجاهل **without replacement** — المقام يتغير في السحبة الثانية.
5. وضع الشيء الخطأ في مقام الاحتمال الشرطي — **الشرط هو المقام**.
6. اعتبار **of those who** احتمالًا عاديًا — هو احتمال شرطي.
7. الخلط بين **at most** و **less than** — كلمة **at** تعني أن الرقم **داخل** الحساب.
8. حساب **at least one** بالطريقة الطويلة — استخدم $1 - P(none)$.
9. استخدام التباديل للجان — اللجنة بدون مناصب = **توافيق**.
10. الجمع بدل الضرب عند الاختيار من فئتين — **and** تعني **ضرب**.
11. كتابة $0! = 0$ — الصحيح $0! = 1$.
12. خلط التباديل والتوافيق بين البسط والمقام.
13. كتابة الناتج النهائي فقط — اكتب فضاء العينة واسم القانون والتعويض.
14. احتمال أكبر من 1 أو سالب — يعني أن في الحل خطأ؛ راجعه قبل أن تنتقل.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability

04 · Lesson 4-4 — Counting Rules

05 · Lesson 4-5 — Probability and Counting Rules

06 · Master formula sheet

07 · Which rule do I use?

08 · The 14 traps that cost marks ← this file

09 · Review exercises, solved

10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.